
A quick prediction of hardness from water quality parameters by artificial neural network

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Abstract: Hardness is an important water quality parameter to determine the suitability of the water for use in different purposes. The hardness of water is conventionally determined by EDTA titration method, which is fairly accurate but time-consuming. Sensor based analyser for hardness, on the other hand, is quite expensive and not easily available. Thus, the conventional methods are inconvenient for the systems, where quick estimation of hardness is essential. This study proposes a model to predict the hardness of water from a few quickly measurable water quality parameters, having high correlations with hardness. The model, developed by artificial neural network, was further validated by a different set of field data. Results indicate that the model is successful in predicting the hardness of water fairly accurately with a high correlation of 0.92, and low deviation (MAPE = 13.60, RMSE = 10.24) of the model predictions from the actual data.

Keywords: artificial neural network; hardness; water quality parameter; WQP; prediction of hardness; tripura; India.

Reference to this paper should be made as follows: Roy, R. and Majumder, M. (2018) 'A quick prediction of hardness from water quality parameters by artificial neural network', *Int. J. Environment and Sustainable Development*, Vol. 17, Nos. 2/3, pp.247–257.

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This paper is a revised and expanded version of a paper entitled 'A quick prediction of hardness from water quality parameters by artificial neural network' presented at 3rd International Conference on Bioenergy, Environment and Sustainable Technologies, Tiruvannamalai, 23–25 January 2017.

1 Introduction

Hardness is an important water quality parameter (WQP) to check the suitability of use in both domestic and industrial purposes (Lower, 2007). Hardness beyond the permissible limits may cause significant health hazards (Kožíšek, 2003) including risks of cardiovascular problems (Pocock et al., 1981; Rubenowitz et al., 1999; Marque et al., 2003) and eczema (McNally et al., 1998; Miyake et al., 2004; Arnedo-Pena et al., 2007). Hard water also causes scale formation on the inner wall of boilers to decrease its efficiency significantly and may also lead to boiler failure (Lower, 2007). In industry, where boilers are used, it is very important therefore to estimate the hardness of water before use.

The conventional and mostly used method of estimating hardness in water is EDTA Titration method (Saravanakumar and Kumar, 2011). This analytical method is time-consuming, troublesome and requires a water quality laboratory setup to perform. In recent years the analytical methods are getting obsolete and the sensor-based water quality monitoring devices are widespread as they are very convenient to use. WQP can be measured by such devices much more easily and quickly, without collecting any sample. The sensor based devices to estimate hardness, however, are expensive, not easily available and also estimate hardness indirectly.

Artificial neural network (ANN) is a computational model inspired by the schematic functioning of the biological neural network. It is an unsupervised machine learning model, which can be trained with a set of input and their output (McClelland et al., 1986). A relation between input(s) and output is developed from iteration modeling by putting adaptive weights in the multiple neural layers (Agatonovic-Kustrin and Beresford, 2000). ANN can be used to predict the output from input(s) when the relation between input and output is not known as it can develop a mathematical relation (a model) between the given set of input(s) and output through regression analysis (Specht, 1991). ANN has been used for forecasting of water quality (Goyal et al., 2015), prediction of some WQP from some other WQP (Khandelwal and Singh, 2005; Swathi and Lokeshappa, 2015) and prediction of hardness (Sirisha et al., 2008).

The present study proposes a quick and easy method to predict hardness of water from a few others easily and rapidly measurable WQP with an ANN model. In this model, hardness can be predicted instantly and enables the managers to take appropriate actions for maintaining the desired level of hardness. The model seems to be very useful in industries, where boiling of water is required. This method can effectively protect the boilers from scale formation without hampering the process flow and without installing expensive monitoring devices.

2 Methodology

A model was developed by ANN to predict the total hardness of water from some other easily measurable WQP. For training of the model, a number of water quality data were collected from the different sources of surface water in Tripura, a North Eastern State of India. The suitable WQP were selected as the predictor of hardness. ANN models were then developed to predict hardness from the selected WQP. Finally, the model was validated by more field data.

2.1 Collection of data

Different WQP of 147 water bodies in Tripura, a north-eastern state of India were estimated with sensor-based multiparameter water quality monitoring devices in the field. Total hardness of the same water bodies were estimated in the laboratory by EDTA Titration method. After the development of the model, the same WQP of another 38 water bodies were estimated for validation of the model.

2.2 Selection of WQP

As the objective of the study is to develop a quick and easy estimation of hardness, only such WQP were measured which can be measured instantly by commonly available inexpensive sensor-based devices. The WQP, having a correlation of ± 0.1 or more with hardness were selected as the predictor WQP for prediction of hardness in this study.

2.3 Development of prediction model

Different ANN models were developed with the selected WQP with combinations of ANN algorithms like combinatorial (COM), stepwise forward selection (SFS), stepwise mixed selection (SMS) and GMDH neural network (GNN) (Table 1).

Table 1 Selection of most important model predictor among the selected WQP

<i>Model</i>	<i>Predictor WQP</i>	<i>Methods used</i>
M1	T, EC, TDS, DO, C	COM, SFS, SMS, GNN
M2	T, EC, TDS, DO	COM, SFS, SMS, GNN
M3	T, EC, TDS	COM, SFS, SMS, GNN
M4	T, EC	COM, SFS, SMS, GNN
M5	T	COM, SFS, SMS, GNN

Notes: T = temperature, EC = electrical conductivity, TDS = total dissolved solids, DO = dissolved oxygen, C = chloride.

In COM method, the model is developed by forming polynomial functions of linear parameters (Anastasakis and Mort, 2001). SFS is a regression analysis where a model is being optimised by inclusion of suitable variables until it becomes statistically significant (Lancashire et al., 2008). SMS, on the other hand, optimises a model by both inclusion

and exclusion of variables (Anastasakis and Mort, 2001). GNN develops a model by iteratively creating layers of neurons where the neurone connections are being optimised by COM algorithm (Anastasakis and Mort, 2001). All these algorithms being automated regression analysis, they can be used to rapid development of optimised ANN models inattentively.

The accuracy of the ANN models were determined by different statistical methods like mean absolute percentage error [equation (1a)], root mean square error [equation (1b)], coefficient of determination [equation (1c)] and correlation coefficient [equation (1d)] of the predictions with actual data.

Equation (1): Statistical measures for determining the accuracy of the ANN models

$$\text{Mean Absolute Percentage Error (MAPE)} = \frac{100}{n} \sum_{i=1}^n \left| \frac{A_i - P_i}{A_i} \right| \quad (1a)$$

$$\text{Root Mean Square Error (RMSE)} = \sqrt{\frac{\sum_{i=1}^n (P_i - A_i)^2}{n}} \quad (1b)$$

$$\text{Coefficient of Determination (R}^2\text{)} = 1 - \frac{\sum_{i=1}^n (A_i - P_i)^2}{\sum_{i=1}^n (A_i - \bar{A})^2} \quad (1c)$$

$$\text{Correlation Coefficient (r)} = \frac{\sum_{i=1}^n (A_i - \bar{A})(P_i - \bar{P})}{\sqrt{\sum_{i=1}^n (A_i - \bar{A})^2 \sum_{i=1}^n (P_i - \bar{P})^2}} \quad (1d)$$

where

$$\bar{A} = \frac{1}{n} \sum_{i=1}^n A_i \quad \bar{P} = \frac{1}{n} \sum_{i=1}^n P_i$$

n number of samples

A actual values

P predicted values.

The model having the highest correlation between actual and predicted values was selected as the effective model for the prediction of hardness from the other WQP.

2.4 Validation of models

A further 38 water bodies were sampled to estimate the selected WQP for validation of the prediction model. The values of WQP, predicted with the model, were compared with the actual values of hardness to check the accuracy of the prediction models.

3 Results and discussions

3.1 Selection of predictor WQP

Five WQP, viz. temperature (T), dissolved oxygen (DO), chloride (C), electrical conductivity (EC), total dissolved solids (TDS), were selected as predictors from 11 WQP (viz. T, DO, C, EC, TDS, turbidity, pH, nitrate, ORP, ammonia, nitrite) for hardness on the basis of their correlation with hardness (Table 2).

Table 2 Selection of predictor WQP for prediction model on the basis of correlation

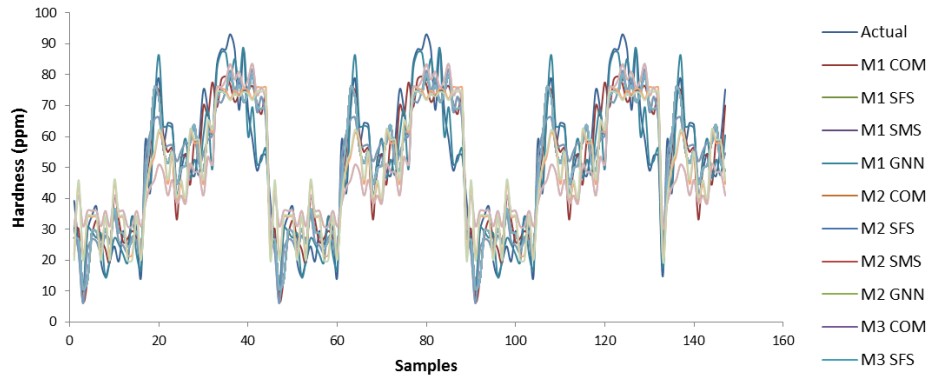
Parameter	<i>T</i>	<i>EC</i>	<i>TDS</i>	<i>DO</i>	<i>C</i>
Correlation	0.76	0.37	0.32	−0.19	0.14

3.2 Development of prediction model

Among the ANN different models developed, M1-GNN seems to be the most accurate (Figure 1) to predict the Hardness for high correlation (0.96) and low errors (Table 3) between actual and predicted set of values.

Table 3 Closeness between actual and predicted hardness for different ANN models during training

Model	Mean absolute percentage error (MAPE)	Root mean square error (RMSE)	Coefficient of determination (R^2)	Correlation coefficient (r)
M1 COM	28.91	23.76	0.55	0.58
M1 SFS	22.37	11.42	0.76	0.87
M1 SMS	22.37	11.42	0.76	0.87
M1 GNN	10.87	6.32	0.93	0.96
M2 COM	22.37	11.42	0.76	0.87
M2 SFS	22.37	11.42	0.76	0.87
M2 SMS	22.37	11.42	0.76	0.87
M2 GNN	21.86	10.94	0.78	0.88
M3 COM	22.37	11.42	0.76	0.87
M3 SFS	22.37	11.42	0.76	0.87
M3 SMS	22.37	11.42	0.76	0.87
M3 GNN	21.86	10.94	0.78	0.88
M4 COM	22.37	11.42	0.76	0.87
M4 SFS	22.37	11.42	0.76	0.87
M4 SMS	22.37	11.42	0.76	0.87
M4 GNN	22.37	11.42	0.76	0.87
M5 COM	28.45	13.60	0.65	0.81
M5 SFS	35.24	15.06	0.58	0.76
M5 SMS	35.24	15.06	0.58	0.76
M5 GNN	27.82	13.53	0.66	0.81

Figure 1 Comparison between model fit values and actual values of hardness during training (see online version for colours)

Thus, the M1 GNN model [equation (2)] can be selected for prediction of the hardness of water from the selected WQP.

Equation (2): The M1-GNN model to predict hardness

$$\text{Hardness} = 2.47497 - 0.0598084D^2 + 0.994685N2 \quad (2)$$

where

T temperature

E electrical conductivity

S total dissolved solids

D dissolved oxygen

C chloride

$$N2 = 2.16581 - N43^2 * 0.00284161 + N3 * 0.898205 + N3^2 * 0.00377312$$

$$N3 = -0.518841 + N46 * 0.141013 + N4 * 0.869328$$

$$N4 = 1.3481 - N33 * 3.03759 + N33^2 * 0.0238478 + N8 * 3.96464 - N8^2 * 0.0229776$$

$$N8 = 0.134989 + N11 * N16 * 0.137534 - N11^2 * 0.0648781 + N16 * 1.01043 - N16^2 * 0.0723653$$

$$N16 = 13.0584 - N365 * 0.680757 - N365 * N22 * 0.00750699 + N365^2 * 0.011468 + N22 * 0.998649 + N22^2 * 0.00342577$$

$$N22 = -8.9104 + N152 * 1.60091 + N152 * N50 * 0.588786 - N152^2 * 0.321544 - N50^2 * 0.273402$$

$$N152 = -1.24444 + N261 * 0.361261 + N280 * 0.663542$$

$$N261 = -1.52655 + N293 * 0.606476 + N312 * 0.423951$$

$$N293 = 56.4392 + N316 * 2.71069 - N316 * N395 * 0.0357128 - N395 * 4.05038 + N395^2 * 0.0591185$$

- N365 $16.7368 + N385 * N386 * 0.0127567$
- N11 $-6.29221 + N41 * 1.59966 - N41 * N24 * 0.164813 + N41^2 * 0.0605948$
 $- N24 * 0.345738 + N24^2 * 0.101396$
- N24 $4.32489 + N43 * N55 * 0.205564 - 43^2 * 0.0998438 + N55 * 0.846029$
 $- N55^2 * 0.103839$
- N55 $0.298769 + N134 * 0.955496 - N134 * N82 * 0.0113419 + N82^2 * 0.0119086$
- N41 $-4.15167 - N82 * 1.00782 - N82 * N125 * 0.0679999 + N82^2 * 0.0475381$
 $+ N125 * 2.13425 + N125^2 * 0.0194067$
- N125 $2.82167 - N359 * 0.417986 + N136 * 1.36175$
- N136 $-1.22061 + N260 * 0.372619 + N280 * 0.651709$
- N260 $-6.61608 + N312 * 1.24855 - N312 * N328 * 0.0126 + N328^2 * 0.0104641$
- N359 $-8.39116 + N361 * 0.958418 + N399^2 * 0.00405347$
- N399 $-3.56576e-11 + N401 * 1$
- N401 $-4.08389 + S * 0.428743 - S^2 * 0.000653731$
- N361 $6.21959 + N363 * N400 * 0.0171292$
- N33 $-8.59019 + N154 * 1.843 + N154 * N50 * 0.598052 - N154^2 * 0.327939$
 $- N50 * 0.256942 - N50^2 * 0.276146$
- N50 $7.65299 + N82^2 * 0.00433117 + N93 * 0.588564$
- N93 $-2.00706 - N282 * N134 * 0.072403 + N282^2 * 0.0379922 + N134 * 1.00405$
 $+ N134^2 * 0.0344082$
- N134 $-5.51618 + N351 * N280 * 0.117869 - N351^2 * 0.0645392 + N280 * 1.46158$
 $- N280^2 * 0.0585664$
- N351 $5.71283 + N357 * N400 * 0.017235$
- N282 $-1.51118 + N305 * 0.651468 + N356 * 0.378652$
- N356 $3.16114 + N363 * 1.1239 - N363 * N386 * 0.0219436 + N363^2$
 $* 0.00887435 - N386 * 0.947152 + N386^2 * 0.0261606$
- N386 $-16.3391 + S * C * 0.0225719 - S^2 * 0.000968765 + C * 4.79886$
 $- C^2 * 0.1851$
- N305 $13.2296 + N316^2 * 0.00905257 + N385^2 * 0.00383793$
- N154 $-1.29856 + N268 * 0.366267 + N280 * 0.659615$
- N268 $-1.47853 + N302 * 0.612313 + N312 * 0.417156$
- N312 $-158.208 + T * 10.471 - T * N385 * 0.107453 + N385^2 * 0.032955$
- N46 $-1.96679 + N76 * 0.969942 - N76 * N102 * 0.27298 + N76^2 * 0.129873$
 $+ N102^2 * 0.143373$
- N102 $71.1893 + T * N243 * 0.115896 - T^2 * 0.164189 - N243 * 1.42597$

- N243 $2.87243 - N387 * 2.14853 - N387 * N287 * 0.0446924 + N387^2 * 0.0435966 + N287 * 3.13363$
- N287 $-14.3779 + N326 * 1.39582 - N326 * N348 * 0.244694 + N326^2 * 0.119939 + N348^2 * 0.121496$
- N76 $2.30665 + N168 * 0.819033 - N168 * N235 * 0.395065 + N168^2 * 0.19966 + N235^2 * 0.196817$
- N235 $7.14497 + N255^2 * 0.00444095 + N276 * 0.594861$
- N276 $-9.73319 + N353 * 0.916052 - N353 * N366 * 0.0248374 + N353^2 * 0.0122445 + N366^2 * 0.0161252$
- N366 $12.5247 + N385 * N400 * 0.0146444$
- N255 $61.7948 + N297 * 2.5126 - N297 * N396 * 0.0307288 - N396 * 3.9973 + N396^2 * 0.0549591$
- N396 $47355 + E * 0.327924 - E^2 * 0.000476036$
- N297 $-10.5505 + C * 2.10923 + C * N316 * 0.0290755 - C^2 * 0.0643087 + N316 * 0.30057$
- N316 $-102.679 + T * 5.22782 + E * 0.149721$
- N43 $2.03468 + N363 * 0.964321 + N363 * N82 * 0.0263952 - N363^2 * 0.0266868$
- N82 $2.07113 - N168 * N169 * 0.348574 + N168^2 * 0.184502 + N169 * 0.800345 + N169^2 * 0.165992$
- N169 $0.044267 - N262 * N280 * 0.0539282 + N262^2 * 0.030993 + N280 * 0.909537 + N280^2 * 0.0241017$
- N280 $11.7586 + N353 * 1.89197 - N353 * N387 * 0.0517259 + N353^2 * 0.0158005 - N387 * 1.91833 + N387^2 * 0.0483589$
- N353 $-5.42344 + N357 * N363 * 0.0227844 + N363 * 1.24149 - N363^2 * 0.025106$
- N262 $-3.32979 + N302 * 1.12243 - N302 * N328 * 0.0103157 + N328^2 * 0.00928528$
- N328 $-9.22481 + C * 2.4635 + C * N352 * 0.0421574 - C^2 * 0.0812158$
- N352 $-111.765 + T * 5.08266 + S * 0.279373 - S^2 * 0.000273135$
- N302 $-175.775 + T * 11.2192 - T * N395 * 0.123045 + N395^2 * 0.0405603$
- N168 $-4.25651 + N263 * 1.09211 - N263 * N300 * 0.12905 + N263^2 * 0.0565825 + N300^2 * 0.0716596$
- N300 $19.7227 + N326 * 3.13077 - N326 * N387 * 0.0440319 - N387 * 2.83114 + N387^2 * 0.0495127$
- N326 $8.29053 + N357^2 * 0.00878098 + N385^2 * 0.00608124$
- N385 $8.94534 + E * 0.395491 + E * C * 0.0301382 - E^2 * 0.00235042 - C^2 * 0.132784$

$$\begin{aligned}
\text{N263} & -20.8974 + \text{N348} * 2.97405 - \text{N348} * \text{N387} * 0.0630208 + \text{N348}^2 \\
& * 0.0106622 - \text{N387} * 1.48154 + \text{N387}^2 * 0.0486102 \\
\text{N387} & 13.3084 + \text{N395} * \text{N400} * 0.014391 \\
\text{N400} & -33.7662 - \text{D} * \text{C} * 0.36803 + \text{D}^2 * 0.533586 + \text{C} * 7.41243 \\
& - \text{C}^2 * 0.0984454 \\
\text{N395} & 7.47355 + \text{E} * 0.327924 - \text{E}^2 * 0.000476036 \\
\text{N348} & 21.3799 + \text{C} * \text{N357} * 0.0437656 - \text{C}^2 * 0.0396217 \\
\text{N357} & -405.324 + \text{T} * 31.1767 + \text{T} * \text{C} * 0.0318839 - \text{T}^2 * 0.527834 \\
\text{N363} & -552.071 + \text{T} * 44.3413 - \text{T}^2 * 0.782628.
\end{aligned}$$

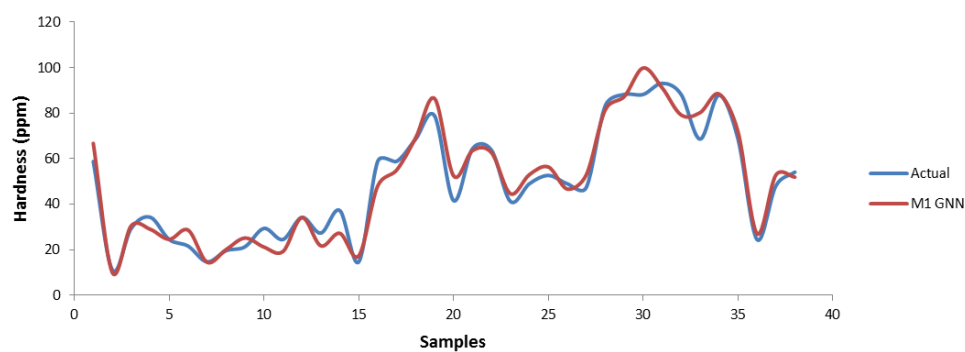
3.3 Validation of model

The M1 GNN model was further validated with more field data. Selected WQP of 38 more water bodies were measured along with hardness. The predicted hardness values were found to be very close to the actual hardness values (Figure 2) with a high correlation of 0.92 (Table 4). Though the errors are significant, but the model seems to predict the approximate hardness fairly well. As Hardness scale is not very sensitive, this model can be used to predict hardness for all practical purposes.

Table 4 Closeness between actual and predicted hardness for different ANN models during validation

Model	Mean absolute percentage error (MAPE)	Root mean square error (RMSE)	Coefficient of determination (R^2)	Correlation (r)
M1 GNN	13.60	10.24	0.82	0.92

Figure 2 Comparison between actual and predicted hardness by M1 GNN model during validation (see online version for colours)



Surprisingly, the T is found to be the most important predictor WQP of this model. Though this fact is interesting, but it is beyond the scope of the present study.

As the predictor parameters can be measured easily and quickly, this model can be used as an easy and quick (though approximate) method to predict the hardness of water.

However, as the model was developed from field data, the model and even the predictors may get changed for a different set of data. Also, this model cannot replace the routine monitoring of hardness through the conventional EDTA method. This model is applicable only for the situations where quick prediction of hardness is essential but accuracy is not very important. Even in such cases, the model should be calibrated regularly with the actual hardness values.

4 Conclusions

An ANN model has been developed to predict hardness of water from T, EC, TDS, DO and C using GNN. The model has been tested against actual field data and found to be fairly accurate in prediction of hardness for all practical purposes. The predicted values have high correlation and low errors with the actual field data. Hence, the model is successful in prediction of the hardness of water from the selected WQP.

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